

# CONCEPT NOTE

## India's Silent Epidemic: A Data-Driven Analysis of Suicide Patterns (SDG 3: Good Health and Well Being)

- **Concept of the Project**

Suicide is a critical public health issue in India, with significant socio-economic and demographic implications. This project aims to analyse and predict suicide rates in India from 2001 to 2012 using machine learning techniques. By leveraging data analysis tools, the project seeks to inform evidence-based prevention strategies and propose actionable interventions to reduce suicide rates for the overall well-being of the population.

- **Problem Statement**

Suicides in India pose a severe challenge, exacerbated by factors such as mental health issues, economic distress, and societal pressures. Despite various efforts, suicide rates remain high due to the lack of precise predictive models and targeted intervention strategies. This project seeks to address this issue by analyzing historical data and developing a machine learning model to predict suicide trends for understanding the complex factors influencing suicide.

- **Objective of the project**

The primary objective of this project is to conduct a comprehensive analysis of suicide data in India to identify patterns, trends, and risk factors associated with suicide. Specific objectives include:

- To analyse historical suicide data from 2001 to 2012 in India.
- To identify key factors like demographic, geographic, and socio-economic of suicide rates.
- To analyse temporal trends in suicide rates to identify potential risk periods.
- To develop a predictive model with an accuracy of 0.82 to forecast future suicide trends.
- To provide evidence-based prevention strategies and recommendations for policy makers and public health officials.

- **Data sources Used**

The project will utilize datasets from the following sources:

- Dataset: Suicides in India 2001-2012.csv
- Sources: - Government health departments
  - Public records
  - Research institutions.

- **Features**

Key features of the dataset will include:

- Demographic information ( Age\_group, Gender )
- Geographic location ( State )
- Circumstances leading to suicide ( Type – i.g. family problems, financial issues, mental health )
- Type\_code
- Temporal data ( Year )
- Total ( suicides )

- **Tools For Analysis**

The following tools will be used for data analysis and visualization:

- **Python:** For data preprocessing, analysis, and visualization.
- **Jupyter Notebooks:** For documenting the analysis process.
- **Scikit-learn:** For machine learning model development.
- **XGBoost:** For advanced gradient boosting models.
- **HistGradientBoostingClassifier:** For robust classification models.

- **Hypothesis**

The hypothesis of the project is that specific socio-economic and demographic factors can significantly predict suicide rates, and targeted interventions based on these predictions can effectively reduce the incidence of suicides.

- **Methodology**

#### **Data Collection and Preprocessing**

- Filter out irrelevant records and encode categorical variables.
- Bin the 'Total' values into categories for better prediction.

#### **Feature Engineering**

- Create new features to capture interactions between variables.
- Scale numerical features for model compatibility.

#### **Model Development**

- Use Gradient Boosting Machine (GBM) and XGBoost for initial model building.
- Perform feature selection using Recursive Feature Elimination (RFE).
- Tune hyperparameters using RandomizedSearchCV for optimal performance.

#### **Model Evaluation**

- Split data into training and testing sets.
- Evaluate models using accuracy, classification reports, and confusion matrices.

#### **Ensemble Method**

- Combine GBM and XGBoost using a soft voting classifier for improved accuracy.

- **Probable Outcome**

The project is expected to:

- A predictive model with an accuracy of 0.82.
- Identification of significant predictors of suicides.
- Data-driven recommendations for targeted interventions.
- Comprehensive analysis reports and visualizations for stakeholders.

This project will provide valuable insights for the development of effective suicide prevention strategies in India, thereby contributing to better public health outcomes.